GMAPS

Ground Movement Awareness and Prediction System

What if the conflict was visible before the call to stop?

LaGuardia Airport · March 22, 2026



● LGA TOWER · MARCH 2026

“Stop Truck 1, STOP.”

Words from the public ATC recording · NTSB investigation ongoing

Dell × NVIDIA Hackathon · Boston

Investigation ongoing · NTSB DCA26MA161

Meet a tower controller.

1 Normally, different jobs on different frequencies.

Local (Tower)	Ground	Clearance Delivery	Controller in Charge
runways · landings & takeoffs · tower frequency	taxiways · aircraft & vehicles · ground frequency	route clearances before taxi	oversees safety of the cab

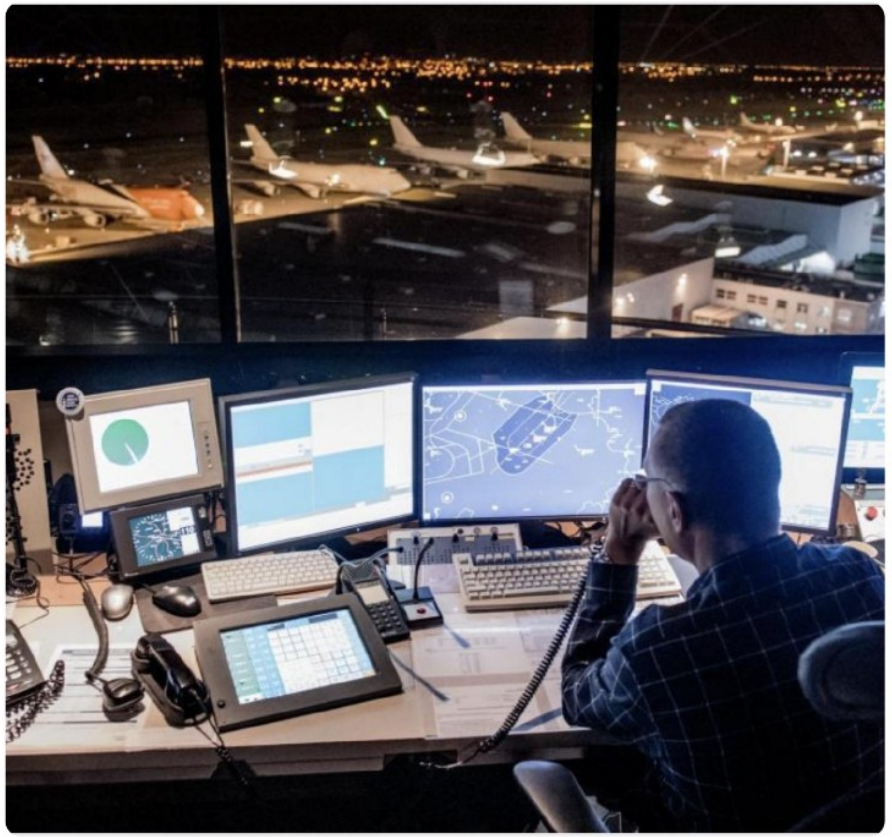
2 When staffing is thin, one person covers several jobs.

2 controllers — **LaGuardia, March 2026:** two controllers covered local, clearance delivery and controller-in-charge duties; NTSB is still determining who was working ground.

3 Either way, no shared picture, no early warning.

Split positions: a pilot on the tower frequency may never hear what is said on the ground frequency.

Combined positions: the whole picture lives in one head.



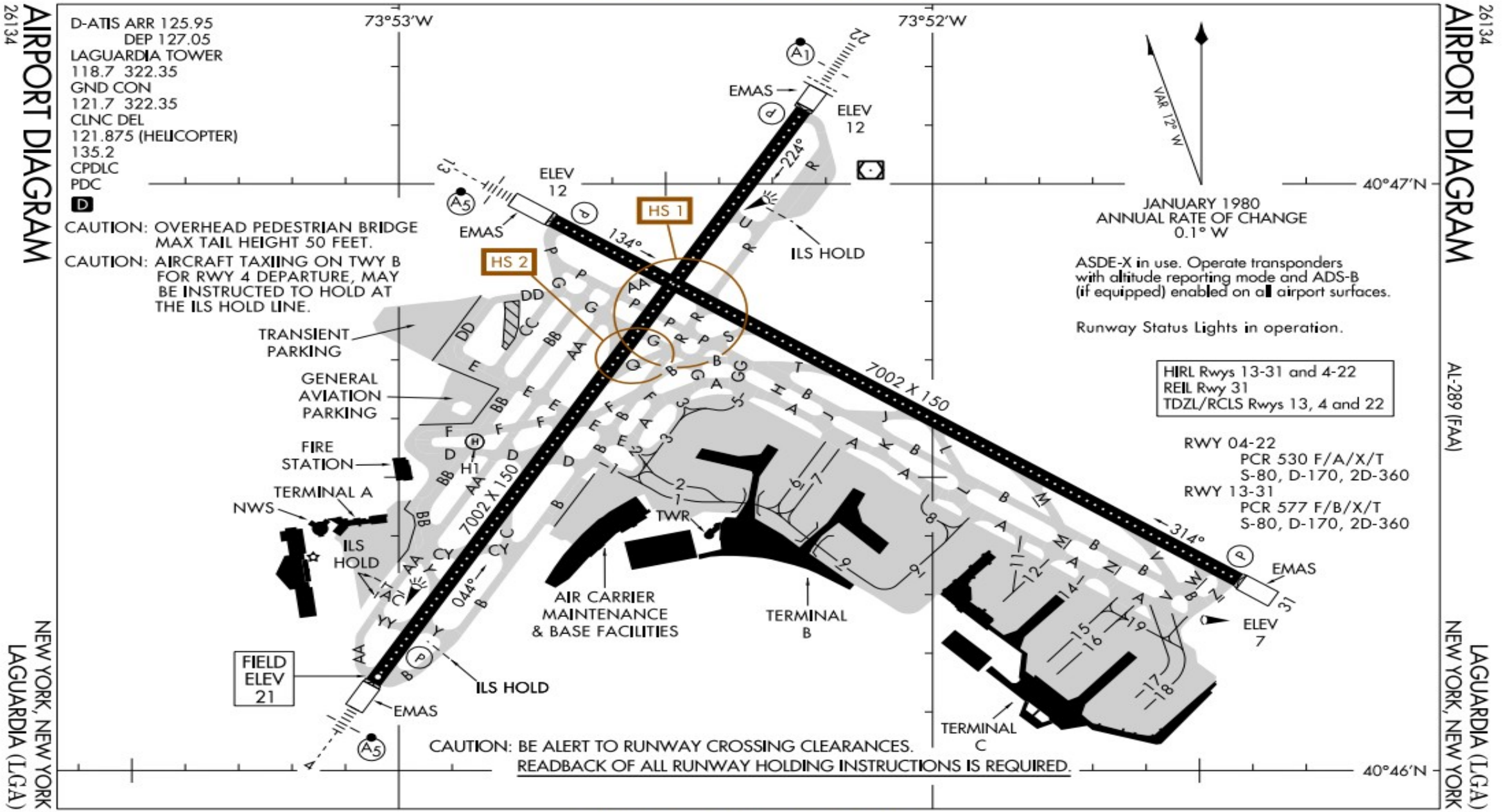
Nothing tracks who is cleared onto which runway, at the same time, across every frequency.

Sources: NTSB investigation as reported by NPR (Mar 24, 2026) and Fox News; the investigation is ongoing. Position descriptions simplified.

As- Is Process



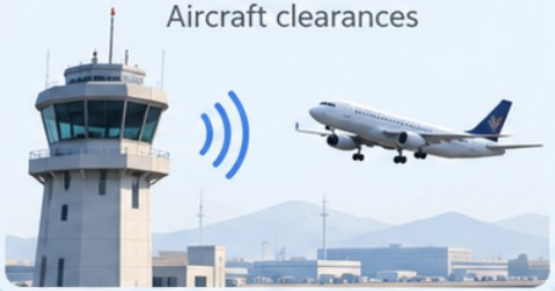



D-ATIS ARR 125.95
DEP 127.05
LAGUARDIA TOWER
118.7 322.35
GND CON
121.7 322.35
CLNC DEL
121.875 (HELICOPTER)
135.2
CPDLC
PDC
D
CAUTION: OVERHEAD PEDESTRIAN BRIDGE MAX TAIL HEIGHT 50 FEET.
CAUTION: AIRCRAFT TAXIING ON TWY B FOR RWY 4 DEPARTURE, MAY BE INSTRUCTED TO HOLD AT THE ILS HOLD LINE.
73°53'W
73°52'W
40°47'N
40°46'N
HS 1
HS 2
ELEV 12
EMAS
ILS HOLD
TRANSIENT PARKING
GENERAL AVIATION PARKING
FIRE STATION
TERMINAL A
NWS
ILS HOLD
FIELD ELEV 21
7002 X 150
7002 X 150
044°
134°
224°
314°
TWR
AIR CARRIER MAINTENANCE & BASE FACILITIES
TERMINAL B
TERMINAL C
ELEV 7
JANUARY 1980
ANNUAL RATE OF CHANGE
0.1° W
VAR 12° W
ASDE-X in use. Operate transponders with altitude reporting mode and ADS-B (if equipped) enabled on all airport surfaces.
Runway Status Lights in operation.
HIRL Rwys 13-31 and 4-22
REIL Rwy 31
TDZL/RCLS Rwys 13, 4 and 22
RWY 04-22
PCR 530 F/A/X/T
S-80, D-170, 2D-360
RWY 13-31
PCR 577 F/B/X/T
S-80, D-170, 2D-360
CAUTION: BE ALERT TO RUNWAY CROSSING CLEARANCES.
READBACK OF ALL RUNWAY HOLDING INSTRUCTIONS IS REQUIRED.

Solution

Tower frequency

Aircraft clearances



Ground frequency

Vehicle movements and crossing clearances



Simulated movement

Aircraft + ground vehicles (SIM-1)



Airport map context

Runway / taxiway layout (fictional training map)





Single local point of processing

Dell Pro Max · NVIDIA GB10

No cloud model. Runs entirely on-site.

Structured intent

Who, what, where, when



Ground state

Current and near-term positions and clearances



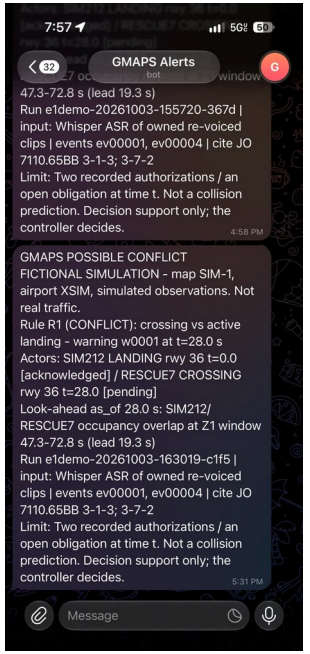
Conflict look-ahead

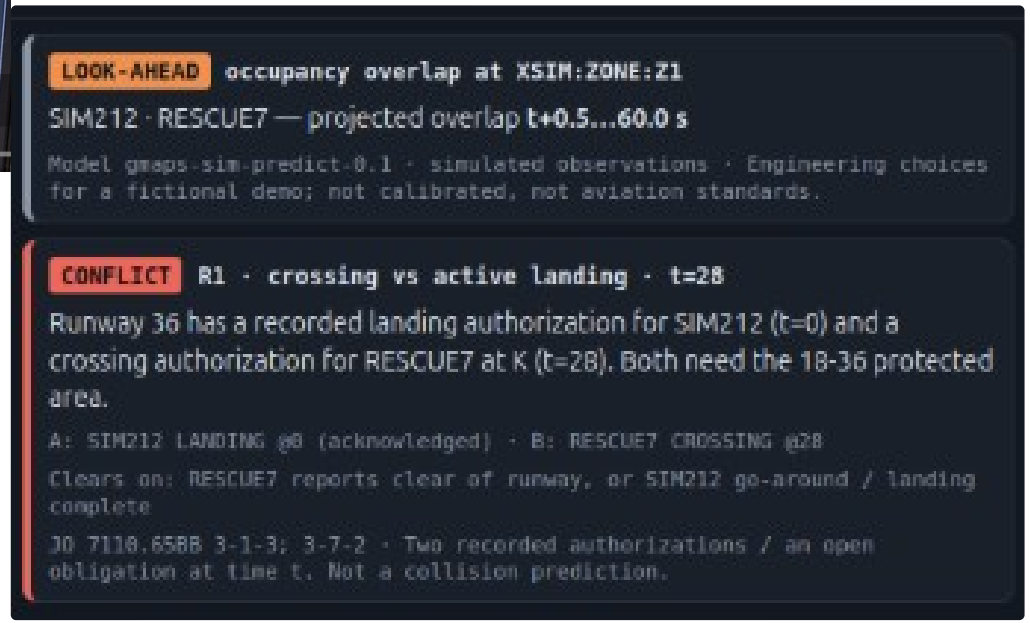
Predicts potential runway-zone overlap

Best placement: Slide 4 or the right half of As-is vs To-be


GMAPS
e1demo-20261003-163019-CTS · input:
Whisper ASR of owned re-voiced clips
FREQUENCY TRANSCRIPT · SPEECH ACT
+0s tower instruction LANDING_CLEARANCE SIM212 · input: whisper
Simair 212, Runway 36, cleared to land.
▸ agent summary (unverified)
+3s tower readback LANDING_CLEARANCE · input: whisper
cleared to land runway 36?R212
▸ agent summary (unverified)
+24s tower request CROSS_REQUEST · input: whisper
Tour, Rescu 7, request to cross runway 36 at Kilo.
▸ agent summary (unverified)
+28s tower instruction CROSS RESCUE7 · input: whisper
Rescue 7, Cross Runway 36 at Kilo.
▸ agent summary (unverified)
+31s tower readback CROSS · input: whisper
Crossing Runway 36 at Kilo, RESCU 7.
▸ agent summary (unverified)
LOOK-AHEAD · FICTIONAL SIM-1 MAP (NOT A REAL AIRPORT)
predicted overlap 50–76 s (model)
Z1
RWY 36 → (fictional)
SIM212
hold
RESCUE7
projected: enters Z1 in ~19s (model) · start time assumed
TWY K
aircraft obs · vehicle obs · hold line · red box = zone Z1
solid trail = observed history; dashed = model projection (simulated, not truth). Positions are simulated observations, frame: SIM-LOCAL-1 (fictional metres).
zone Z1 occupancy windows, now = +60s (combined = intent + motion)
AC combined
VH combined
Pause
t = +31 s
WARNINGS · EXPLAINABLE
LOOK-AHEAD occupancy overlap at XSIM:ZONE:Z1
SIM212 · RESCUE7 — projected overlap t+19.2…44.7 s
Model gmaps-sim-predict-0.1 · simulated observations · Engineering choices for a fictional demo; not calibrated, not aviation standards.
POSSIBLE CONFLICT alert sug 18
CONFLICT R1 · crossing vs active landing · t=28
Runway 36 has a recorded landing authorization for SIM212 (t=0.0) and a crossing authorization for RESCUE7 at K (t=28.0). Both need the 18-36 protected area.
A: SIM212 LANDING @0 (acknowledged) · B: RESCUE7 CROSSING @28
Clears on: RESCUE7 reports clear of runway, or SIM212 go-around / landing complete
JO 7110.65BB 3-1-3; 3-7-2 · Two recorded authorizations / an open obligation at time t. Not a collision prediction.
STATE LEDGER (FIVE KINDS KEPT SEPARATE)
actor
authorized
acknowledged
reported
inferred
observed
SIM212
LANDING_CLEARANCE @0
–
–
–
(902, -1) m, age 1s [simulated]
RESCUE7
CROSS @28
–
–
–
(1504, -302) m, age 1s [simulated]







Live prototype · fictional exercise · simulated movement observations

http://localhost:8765/index.html?f=samples/LGA_2026-03-22.json

GMAPS HISTORICAL REPLAY · RE-VOICE · SECONDARY SOURCES — LaGuardia, Mar 22, 2026 — crossing clearance issued on runway 4 with a landing clearance active

LGA_2026-03-22 | No file selected.

FREQUENCY TRANSCRIPT · SPEECH ACT

2384, we have an odor on the plane as well at this time. We are going to be going back to the gate, request fire as well.

-259s tower unknown OTHER
Okay. Do you know what gate? Is this odor like a smoke odor, like a fire?

-249s tower unknown OTHER
No, it was a weird odor. We can't get a hold of anyone at ops for a gate assignment.

-233s phone PHONE PHONE
Tower calls the ramp and the Port Authority about a gate (several calls, video 5:57–6:55).

-165s tower unknown OTHER ?2384
2384, I just got off the phone with your ramp, they don't have a gate for you. Is this still an emergency? Do you still need us to send fire?

-148s tower unknown OTHER
Yeah, there's still an alert in the back.

-138s tower unknown OTHER ?2384
2384, do you plan on evacuating at all?

-136s tower unknown OTHER
Negative.

-122s tower instruction TAXI ?2384
2384, make the left turn on Kilo and then right on Alpha, short of Echo.

-120s tower readback TAXI ?2384
Kilo, Alpha, short of Echo, 2384.

-113s tower report REPORT_EMERGENCY ?2384
2384 is declaring an emergency. The flight attendants in the back are feeling ill because of the odor. We will need to go into any available gate at this time.

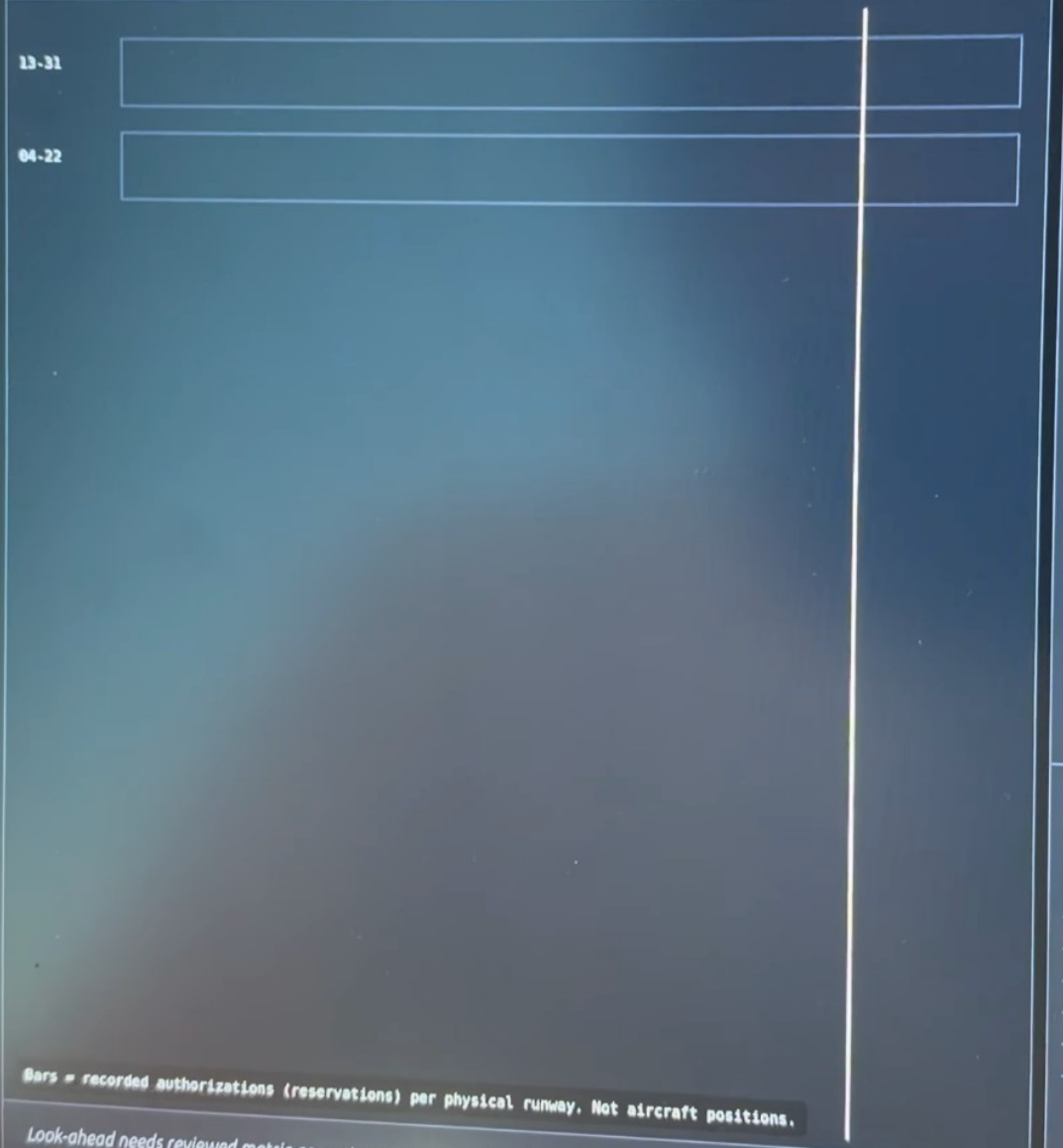
-104s tower unknown OTHER
Okay, roger that.

-70s phone PHONE PHONE
Tower to ramp: no gate; a stair truck is offered. 'He's on taxiway Alpha just east of taxiway Kilo.'

-58s tower unknown OTHER ?2384
2384, the ramp doesn't have a gate for you yet, I'm trying to get in touch with the Port Authority.

-58s tower unknown OTHER
And the fire trucks are over there. They're going to bring a stair truck just in case you want to evacuate. Let me know if you do.

RUNWAY RESERVATIONS (RECORDED AUTHORIZATIONS, NOT POSITIONS)



Bars = recorded authorizations (reservations) per physical runway. Not aircraft positions.

Look-ahead needs reviewed metric geometry + independent tracks; this historical replay has no episodes.

WARNINGS · EXPLAINABLE

No open warnings at this time.

STATE LEDGER (FIVE KINDS KEPT SEPARATE)

actor	authorized	acknowledged	reported	inferred	observed
UAL2384	taxi, via K-A, hold short of E @-122	read back: taxi, via K-A, hold short of E @-120	emergency @-113	exit instruction issued: runway 13 assumed vacating (not observed) @-297	–
JZA640	–	–	–	–	–
DAL520	–	–	–	–	–
TRUCK1	–	–	–	–	–
FPT4198	–	–	–	–	–
DAL263	–	–	–	–	–

Tech Stack

GMAPS · Ground Movement Awareness and Prediction System

ATC Radio Audio

re-voiced fictional tower & ground calls

Simulated Movement

fictional SIM-1 map · timestamped positions

Rules & Phraseology

FAA phraseology grammar · rules R1–R13 (JO 7110.65)

DELL | NVIDIA | Dell Pro Max GB10

LOCAL AI PIPELINE

Speech → Text

Whisper large-v3-turbo

turns radio into text, locally · ~0.2 s per clip

Agent & Parsing

Qwen3.6-35B-A3B (NVFP4) · vLLM

reads each call, chooses tools; extracts callsign, action, runway, taxiway

Ledger & Look-ahead

five-state ledger · rules R1–R13

predicts future runway-zone overlap from intent + movement

REVIEW & EVIDENCE



Review screen · live GB10 telemetry · run log with run ID, transcript, evidence & limits

NemoClaw / OpenShell Orchestration

manages the stack + local vLLM server

OpenShell sandbox

- sandboxed agent runtime
- egress policy: cloud model routes removed
- local inference route (inference.local)

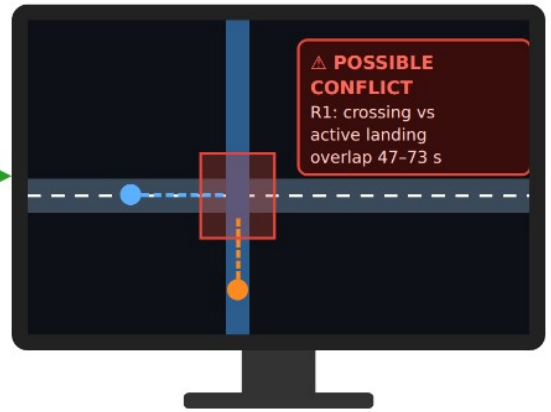
OpenClaw agent

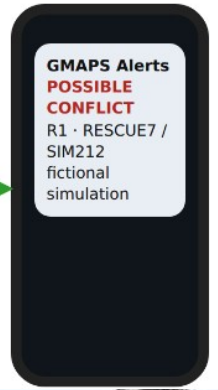
- tools-only agent (no shell, no web)
- 5 GMAPS tools over local MCP

`parse · ingest · look-ahead warnings · post_alert`

LOCAL ON THE GB10 • NO CLOUD MODEL • ONLY THE ALERT TEXT LEAVES THE BOX

Controller Review Screen





Telegram alert to supervisor / instructor

The human decides.



Local AI decision support for airport ground operations

EXISTING INFRASTRUCTURE

35 + 9

FAA lists 35 ASDE-X airports and 9 ASSC locations.
Existing systems provide conflict alerts.

INITIAL USER HYPOTHESIS

Airport ops + ATC

Start with replay and training review.
Validate the workflow and buyer.

BUSINESS MODEL HYPOTHESIS

Paid evaluation

Test pilots, then software + support.
Pricing and market size to validate.

	Phase 1: Validate	Phase 2: Evaluate	Phase 3: Expand
Focus	Replay + simulation	Partner evaluation	Additional airports
Product	Radio + clearances + simulated tracks	Authorized data + shadow-mode review	Repeatable integrations + support
Milestone	Expert review + benchmark results	Data access + measured usefulness	Required approvals + repeatable results

Validate the evidence → Evaluate with users → Expand after proof

Sources: FAA ASDE-X (2023) and ASSC (2024); checked Oct 3, 2026. Counts describe existing FAA systems.
Proposed roadmap. GMAPS market size, pricing and customer adoption remain unvalidated.

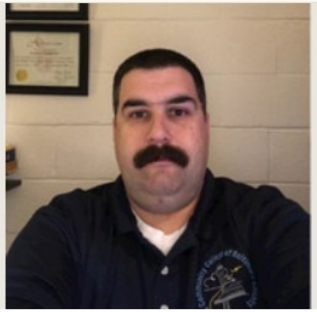


Hear the call.
See the conflict developing.

LaGuardia reminds us why every second matters.

AI connects the evidence. Humans decide.

Thank you to our domain collaborators



Joe Eichelberger

ATC Director, Community College of Baltimore County · retired military controller

Co-designed controller-side state tracking with FAA phraseology as the grammar.



EMBRY-RIDDLE AERONAUTICAL UNIVERSITY

Andrew “Andy” Schneider & Dr. Jianhua Liu

Co-leads, Speech & Language AI Lab · aviation speech recognition · pilot

Their ATC speech research informed our voice pipeline, and they reframed the problem: issued instruction vs. actual compliance.



Capt. Jeff Nielsen

Host, Airline Pilot Guy · retired Delta captain, former USAF pilot

The cockpit view: why a mechanical second listener matters.



Scott Gyssler

Senior Director, ATC Programs, Broward College · former FAA controller

Drill framing: playback teaches, it never replaces judgment.



Angela Taylor

Tower Chief, Middle Georgia State University (Eastman, GA) · 27 years in ATC

Ground truth from the tower: radio-loss events get logged, rarely debriefed, which is the gap review tools can fill.



Dr. Ella Atkins

Head, Aerospace & Ocean Engineering, Virginia Tech · AIAA Fellow

Asked the question that shaped the build: what data do you have access to?

Built with the people who do this work.

Domain collaborators: their expertise shaped this design, including the corrections that made it honest. No collaborator endorses the product.

OPEN AGENT HACKATHON 2026 · WILDCARD [TINKERER]

New this hackathon: an agent loop with roles that check each other

Planner grammar first; what the grammar cannot type goes to Nemotron; look-ahead only for a shared zone.

Executor bounded tools only: parse, ledger, look-ahead, explain, alert. No shell.

Evaluator grounding rule on every model answer; one retry with the airport vocabulary; then escalate "needs readback".

Explainer + Critic Nemotron drafts two sentences; a second Nemotron call critiques them against the facts; one revision, else labelled ledger text.

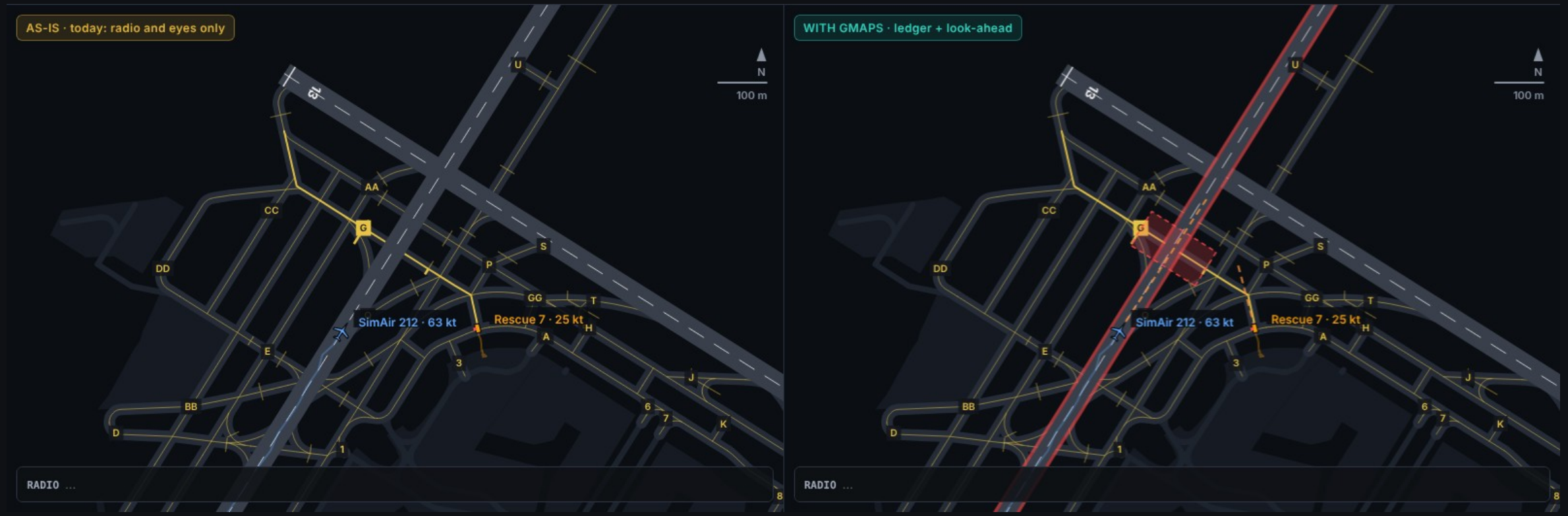
Risk SIM-1 look-ahead; stale positions give "prediction unavailable", not "no conflict".

Traces one JSON line per step: role, provider, model_reported, latency, outcome (logs/, tools/show_trace.py).

Run it python3 run.py · POST /run · SAMPLE_MODE=true with no credentials · 3 input/output examples · 21 new tests passing.

NEW · SIMULATED · FICTIONAL TRAFFIC ON THE REAL KLGA LAYOUT

LaGuardia view: AS-IS vs WITH GMAPS, side by side



HONEST STATUS

Sponsor technology: what is real at submission time

NVIDIA Nemotron: nemotron-3-nano-30b-a3b-nvfp4 served locally by vLLM on the DGX Spark (GB10); it runs the explainer and critic roles. Measured run: 22 Nemotron calls, GPU peak 96%, every connection loopback (runs/proof/local_20261027-233316).

Meterless Nightshift: provider client built (OpenAI-compatible, Console URL and runtime key, meterless headers, retry, typed "unavailable"). Live run through Nightshift: not run (Nightshift was not running at submission time).

Meterless Engine (H-MEM): not integrated. A local runway memory with the same role runs in its place, with a passing test.

Zetaris: not integrated. The table design and an idempotent loader are ready; the agent uses a local store with the same interface.

Local-execution proof: tools/local_proof.py audits every connect() and charts GPU use from nvidia-smi during a run.

DECLARATION

Pre-existing vs new

Pre-existing (commit e97a9a1 and earlier): grammar parser, ledger and rules, SIM-1 simulator and look-ahead, airport data, fine-tuned Qwen3-4B LoRA parser (record exact 0.254 → 0.652 on 138 held-out Whisper-text lines), vLLM / DGX Spark runtime, original demo page.

New (e97a9a1..HEAD): agent loop with critic, Nightshift provider, traces, sample-mode replay, run.py API, LaGuardia animated view, local-execution proof, Nemotron-vs-4B comparison runner, deck and docs.

Limits: simulated traffic and positions, synthetic phraseology; decision support only; GMAPS never transmits, the controller decides.